Important Information

Thank you for choosing Freenove products!

Getting Started

If you have not yet downloaded the zip file associated with this kit, please do so now and unzip it.
https://github.com/Freenove/Freenove_Power_Supply_for_Breadboard

Get Support and Offer Input

Freenove provides free and responsive product and technical support, including but not limited to:

- Product quality issues
- Product use and build issues
- Questions regarding the technology employed in our products for learning and education
- Your input and opinions are always welcome
- We also encourage your ideas and suggestions for new products and product improvements

For any of the above, you may send us an email to:

support@freenove.com

Safety and Precautions

Please follow the following safety precautions when using or storing this product:

- Keep this product out of the reach of children under 6 years old.
- This product should be **used only when there is adult supervision present** as young children lack necessary judgment regarding safety and the consequences of product misuse.
- This product contains small parts and some parts that are sharp. This product contains electrically conductive parts. **Use caution when placing conductive parts near power supplies, batteries, or live circuits.**
- When the product is powered on, some parts will move or rotate. **To avoid injury, keep your hands and fingers away from moving parts.**
- It is possible that an improperly connected or shorted circuit may cause overheating. **If this happens, immediately disconnect the power supply or remove the batteries. Do not touch anything until the product has cooled down!** When everything is safe and cool, review the product tutorial to identify the cause.
- Only operate the product in accordance with the instructions and guidelines of this tutorial, otherwise parts may be damaged or you could be injured.
- Store the product in a cool dry place and avoid exposing the product to direct sunlight.
- After use, always turn the power OFF and remove or unplug the batteries before storing.

About Freenove

Freenove provides open source electronic products and services worldwide.

Freenove is committed to assist customers in their education of robotics, programming and electronic circuits so that they may transform their creative ideas into prototypes and new and innovative products. To this end, our services include but are not limited to:

- Educational and Entertaining Project Kits for Robots, Smart Cars and Drones
- Educational Kits to Learn Robotic Software Systems for Arduino, Raspberry Pi and micro: bit
- Electronic Component Assortments, Electronic Modules and Specialized Tools
- **Product Development and Customization Services**

You can find more about Freenove and get our latest news and updates through our website:

http://www.freenove.com

sale@freenove.com

Copyright

All the files, materials and instructional guides provided are released under Creative Commons Attribution-NonCommercial-ShareAlike 3.0 Unported License. A copy of this license can be found in the folder containing the Tutorial and software files associated with this product.

This means you can use these resource in your own derived works, in part or completely but **NOT for the intent or purpose of commercial use.**

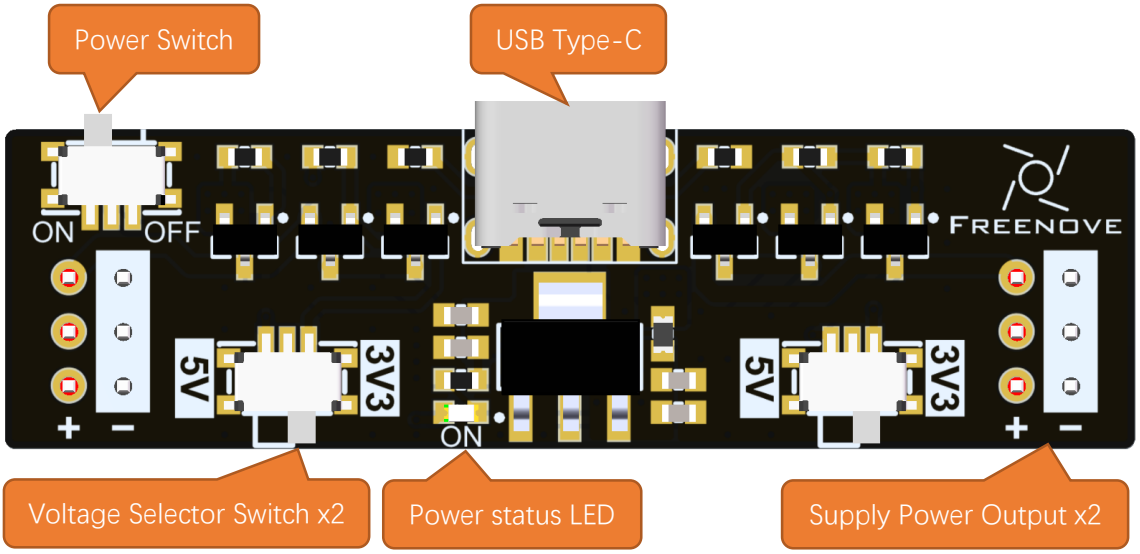
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Instructions for Use

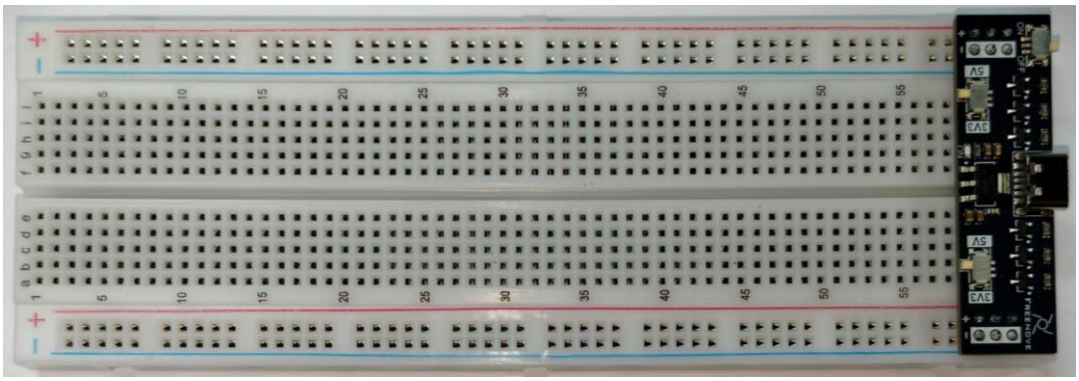
Welcome to use the Freenove Power Supply for Breadboard.



Here are its key features:

- **Breadboard Compatibility**: Compatible with standard MB-102 breadboards (830/850 tie points).
- **Interface Type**: USB Type-C (5V input; supports USB Type-C to Type-C cables).
- **Protection**: Onboard resettable fuse for protection against short circuits and overcurrent.
- Output Channels: Two independent power outputs, each selectable between 5V and 3.3V using the onboard switch.
- **3.3V Voltage Regulation**: Uses an onboard 1117-3.3V linear voltage regulator (LDO). The regulator generates heat during voltage conversion. Higher input-to-output voltage differences and output currents result in greater heat generation. Therefore, the maximum continuous output current is limited by the regulator's thermal dissipation capability.
- **Power Output**: 5V, up to 1A (maximum); 3.3V, up to 0.5A(maximum).
- **Indicator Lights**: Onboard power status LED.

To install the power supply module on a breadboard, refer to the figure below:



If you encounter any issues, feel free to contact us for prompt and free technical support.

support@freenove.com

What's next?

Thank you again for choosing Freenove products.

THANK YOU for participating in this learning experience!

We have reached the end of this Tutorial. If you find errors, omissions, or you have suggestions and/or questions about the Tutorial or component contents of this Kit, please feel free to contact us:
support@freenove.com

We will make every effort to make changes and correct errors as soon as feasibly possible and publish a revised version.

If you want to learn more about Arduino, ESP32, Raspberry Pi Pico W, Raspberry Pi, Smart Cars, Robotics and other interesting products in science and technology, please continue to visit our website. We will continue to launch fun, cost effective, innovative and exciting products.
http://www.freenove.com/

Thank you again for choosing Freenove products.